

Samanvay

Federated interoperability layer · Team fusionX



Problem Statement ID — SIH26129

Problem Statement Title — System integration and interoperability among government digital platforms, resulting in fragmented service delivery

Theme — Miscellaneous

Organisation — Government of Maharashtra

PS Category — Software

Team Name (Registered on portal) — fusionX



How Samanvay addresses the problem

Federated interoperability — not a new portal, not a data lake, not an aggregator. Certificates stay at the issuer.

Departments stay owners

Aaple Sarkar, Mahabhulekh, Fire, MPCB, MahaDBT keep their systems. Samanvay sits between them.

Nothing stored except proof

Links, consent, pointers, workflow, tracking, hash-chained audit. Payloads are fetched live, then discarded.

Three callers, one core

Scholarship, licence/NOC, and farmer subsidy are ready skins on published catalog journeys — no extra Java for farmer.

01

APPLY & LINK

Citizen applies on an existing skin

Scholarship · Licence · Farmer subsidy

Connect department accounts

Map IDs. Probabilistic matches never auto-merge.

DigiLocker may carry issued copies. It is not login and not Samanvay's consent store.

02

CONSENT

Purpose-bound agreement

Requester, purpose, categories — never a blanket dump

Consent artifact

Revoke anytime. Version increments.

A scholarship grant cannot later pull a fire NOC.

03

CONTROL PLANE DECIDES

Access authority

Identity + consent + registry pointer, as one decision

Any no → deny

Reason written on the audit chain

All yes → signed grant

60-second TTL, single-use nonce. No fetch without it.

Control plane decides. Data plane cannot call a department "just this once".

04

DATA PLANE MOVES

Verify grant, then adapt

REST · SOAP · SFTP · JDBC → in-memory canonical map

Live fetch from system of record

REST · SOAP · SFTP · JDBC. Payload never persisted.

If the source is down

Needs action. Officer retries that department. Not a gateway loop — the citizen file is not restarted.

05

TRACK & PROVE

One application number

MH-SCH · MH-NOC · MH-FAR

Officer desk

Which issuer answered. Retry a downed department.

Hash-chained audit

Who asked, when, grant id. Never the certificate.

Revoke → next fetch denied within 60s. Nothing to purge — it was never stored.

Technical approach

Java 21 • Spring Boot 4 • Modulith • Postgres 16 • BPMN 2.0 • Resilience4j • REST • SOAP • SFTP • JDBC • Keycloak planned (AUTH STUBBED in demo)

Scholarship

Revenue • Education • DBT — judge path at /scholarship/

Licence / NOC

Municipal • Fire • Pollution • Revenue — /licence/

Farmer subsidy

Revenue • Agriculture • DBT — /farmer/ • catalog + BPMN, no new Java

Control plane — decides

Identity — citizen links. No auto-merge.

Consent — purpose-bound artifact. Issues the grant. Revoke increments version.

Catalog — departments, schemas, journeys, mappings.

Registry — pointers only: what exists where. Never the payload.

Signed access grant
60s • single-use
typed into runtime



No fetch
without a grant

Data plane — moves

Orchestration — BPMN fan-out, retry, degrade.

Connector — adapter + mapping DSL.

In-memory canonical map — write freshness + audit only.

Tracking — one reference: MH-SCH / MH-NOC / MH-FAR.

Cross-cutting audit — hash-chained ledger. Every grant, fetch, retry, and revoke. Tamper a row → chain verify fails. Live fetch with provenance; certificates stay at the issuer. Inspired by Account Aggregator consent — Samanvay is not an aggregator and does not store papers.

Aaple Sarkar
income / caste

Mahabhulekh
7/12 land

Fire e-approval
fire NOC

MPCB / MAITRI
pollution clearance

MahaDBT
bank / subsidy

DigiLocker*
issued copies, not consent

Path: identity link + consent → signed grant → adapter fetch (not stored) → canonical map → track / officer. *DigiLocker is an optional issued-copy pipe, not department login and not the consent store. Demo uses labelled sandbox + mock adapters.

Feasibility and viability

Feasible now

- Core already runs: Java 21 / Boot 4 / Postgres / three published journeys.
- Heterogeneous mocks are the proof {SOAP, REST, JDBC, SFTP}.
- PS constraint met: no replacement of existing department systems.
- Onboarding a journey is catalog + connectors + BPMN, not a rewrite. Farmer subsidy proves this — zero new Java, ready skin at /farmer/.
- Demo is fully offline — venue network is not a dependency.

Challenges / risks

- Live DigiLocker / Aadhaar needs a MeitY partner account (not on this laptop).
- Real department APIs need MoU / NIC / department credentials.
- Keycloak brokering and BPMN depth are fiddly in production. Demo identity is AUTH STUBBED.
- Probabilistic identity merge is unsafe if auto-applied.
- Officer queues and SLA if a source stays down — not a gateway retry loop.

How we overcome them

- Sandbox labelled as sandbox; issuer mocks stand in for systems of record.
- WorkflowEngine port: BPMN today, YAML DAG fallback in ~2 days.
- CHECK + service + RBAC: a probabilistic match never becomes an active link.
- Degraded mode + officer retry (mark a department unavailable in demo, restore, retry).
- Recorded backup walkthrough if live demo fails.

Citizens

Facts already issued (income, caste, 7/12, marks, land, crop) are fetched live under consent — not re-uploaded, not stored in a lake. One application number. Revoke and the next fetch is denied; nothing to purge.

Officers

One file, not three portals. See which issuer answered. Retry a downed department without restarting the citizen journey. Status in officer words: submitted, in progress, needs action, completed.

Departments

Keep their system of record. Plug in via adapter + mapping. A new scheme is configuration — farmer subsidy already demonstrates this, with a citizen skin on the same core.

State / economic

Avoid a fourth silo or a central data lake. An interoperability layer is cheaper to extend than N×N bilateral integrations.

Social / privacy

Sensitive discovery (for example a caste pointer) is consent-gated. Provenance on every field. Tamper-evident audit for RTI / inquiry. DigiLocker is not treated as consent.

Environment

Fewer duplicate physical certificates and counter visits; digital reuse of already-issued records, fetched on demand.

Problem & policy

- SIH26129 — GoM / MSINS: interoperability without replacing existing systems
- India Stack / DEPA-style consent artefacts (purpose, requester, revoke)
- MeitY DigiLocker issued-document model (issuer push + citizen pull) — copies, not Samanvay consent

Maharashtra systems of record (adapted, not scraped)

- Aaple Sarkar — aaplesarkar.mahaonline.gov.in — income / caste
- Mahabhulekh — bhulekh.mahabhumi.gov.in — 7/12 Record of Rights
- MahaDBT — mahadbtt.maharashtra.gov.in — scholarship / farmer payment
- Fire e-approval — mahafireservice.gov.in/e-fire.php
- MAITRI / MPCB — maitri.maharashtra.gov.in — pollution clearance
- MahaVastu / BPMS — mahavastu.maharashtra.gov.in

Novelty vs alternatives

Approach	Speed	Fit
Point-to-point	Low	Poor — N×N
REST / SOAP only	Medium	Misses SFTP / JDBC
Generic ESB / lake	High	Stores papers
Samanvay (federated layer)	High	Pointers + 60s grant + live fetch

Standards we implement against

- BPMN 2.0 for cross-department journeys
- OIDC / SAML via Keycloak when live (demo: AUTH STUBBED)
- REST, SOAP/XML, SFTP/CSV, JDBC
- JSON Schema canonical model + JSONPath mapping DSL
- Hash-chained audit (tamper-evident ledger)

This prototype

- [docs/architecture/HLD.md](#) and [LLD.md](#)
- [docs/demo/PHASE4_RUNBOOK.md](#) · [JUDGE_SCRIPT.md](#)
- github.com/himanshu2394i/samanvay

Honest demo boundary

- Issuer APIs and DigiLocker partner credentials are mocked / sandboxed until departments and MeitY onboard.
- Samanvay is not an aggregator and does not persist certificates.
- A downed source surfaces as needs action + officer retry — the citizen file is not restarted.